OSCILLATOR

Objectives

After studying the material in this chapter, you should be able to:

1. State the function of the oscillator.
2. Describe *positive feedback,* how it is produced, and how it maintains oscillations after an oscillator is triggered.
3. Discuss the *Barkhausen criterion* and its effect on oscillator operation.
4. List the three requirements for proper oscillator operation.
5. Describe the operating characteristics of the *phase-shift* and *Wien-bridge RC* oscillators.

Objectives

6. Describe the operation and perform the complete frequency analysis of a *Colpitts oscillator*.
7. Describe the operation of the *Hartley, Clapp,* and *Armstrong LC* oscillators.
8. Describe the operating principles of *crystals* and *crystal-controlled oscillators*.

Introduction

An *oscillator* is a circuit that produces an output waveform with only a dc power supply input. The range of frequencies required by electronic devices may range from a few hertz into the megahertz region, and these signals are generally produced by oscillators. We will focus on sine-wave oscillators.

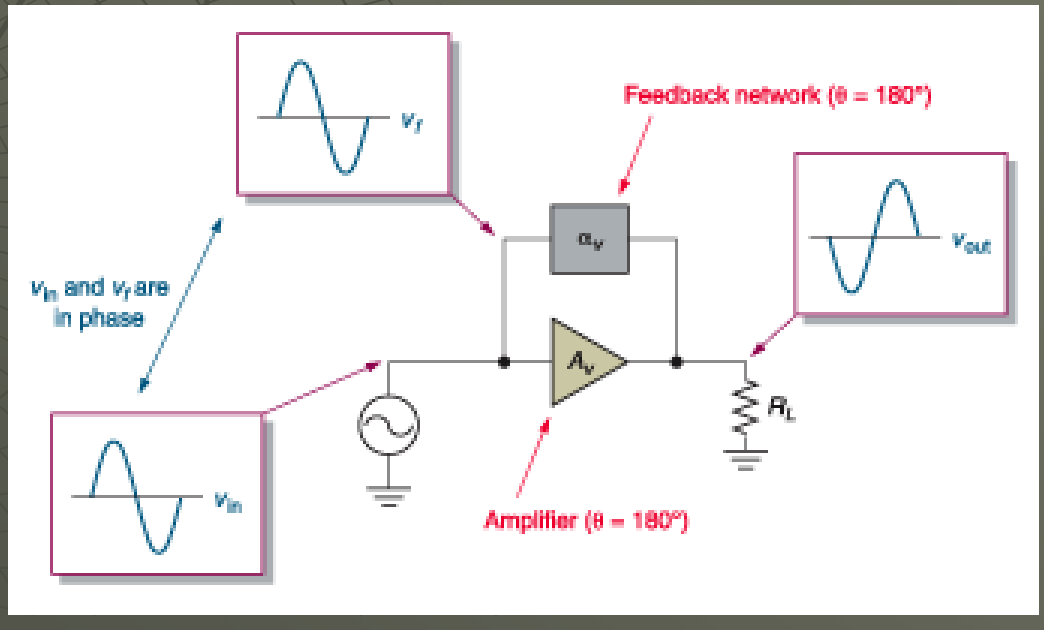
Oscillator

An *oscillator* is a circuit designed to generate an output waveform without an external signal source. The key to oscillator operation is *positive feedback*. A positive feedback network produces a feedback voltage (v_f) that is in phase with the input signal (v_{in}). (Positive feedback can also be obtained using an amplifier and feedback network that both produce a 0° phase shift.) Positive feedback is also referred to as *regenerative feedback*.

Oscillator Operation

An oscillator needs a brief *trigger signal* to start the oscillations. Most oscillators provide their own trigger simply by turning the circuit on. Basic requirements for oscillator operation:

1. The circuit must have *regenerative feedback*; that is, feedback that results in a combined 360°(or 0°) voltage phase shift around the circuit loop.
2. The circuit must receive some *trigger signal* to start the oscillations.



Barkhausen Criterion

There is one other requirement for oscillator operation. The circuit must fulfill a condition referred to as the *Barkhausen criterion*.

The active component in a feedback amplifier produces a voltage gain (A_v) while the feedback network introduces a loss or attenuation (β_v). In order for an oscillator to work properly, the following relationship must be met:

$$\beta_v A_v = 1$$

This relationship is called the *Barkhausen criterion*.

Barkhausen Criterion

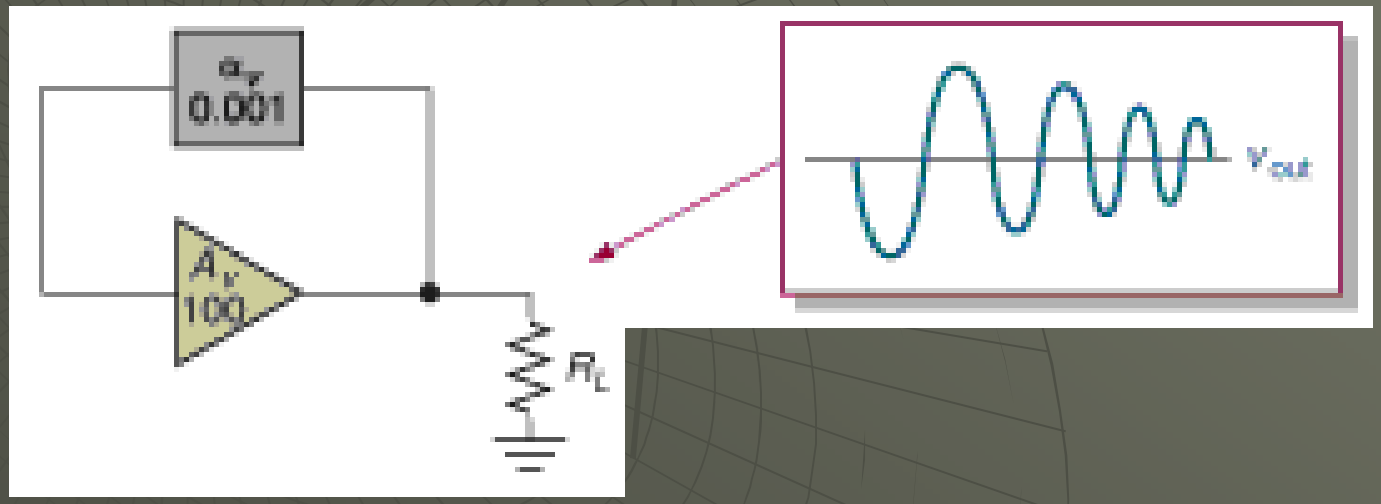
If *Barkhausen criterion* is not met, one of the following occurs:

1. If $\beta_v A_v < 1$, the oscillations die out after a few cycles.

2. If $\beta_v A_v > 1$, the oscillator drives itself into saturation and cutoff clipping.

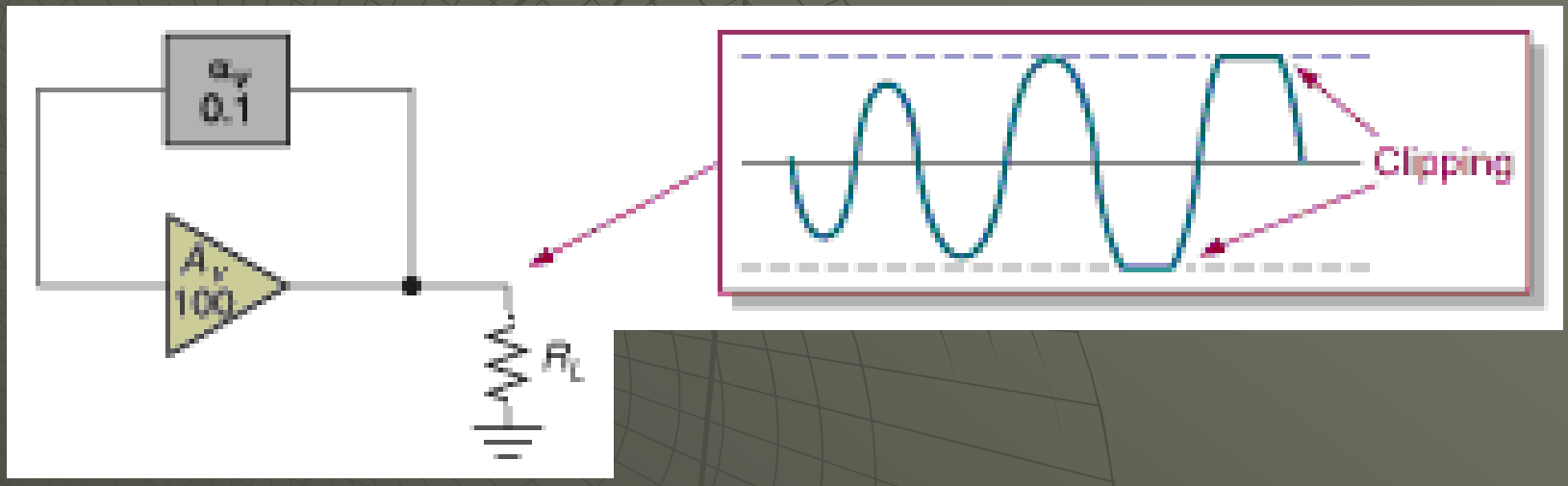
Effects of $\beta_v A_v$

If $\beta_v A_v < 1$, each oscillation results in a lower-amplitude signal being fed back to the input . After a few cycles, the signal fades out. This loss of signal amplitude is called *damping*.



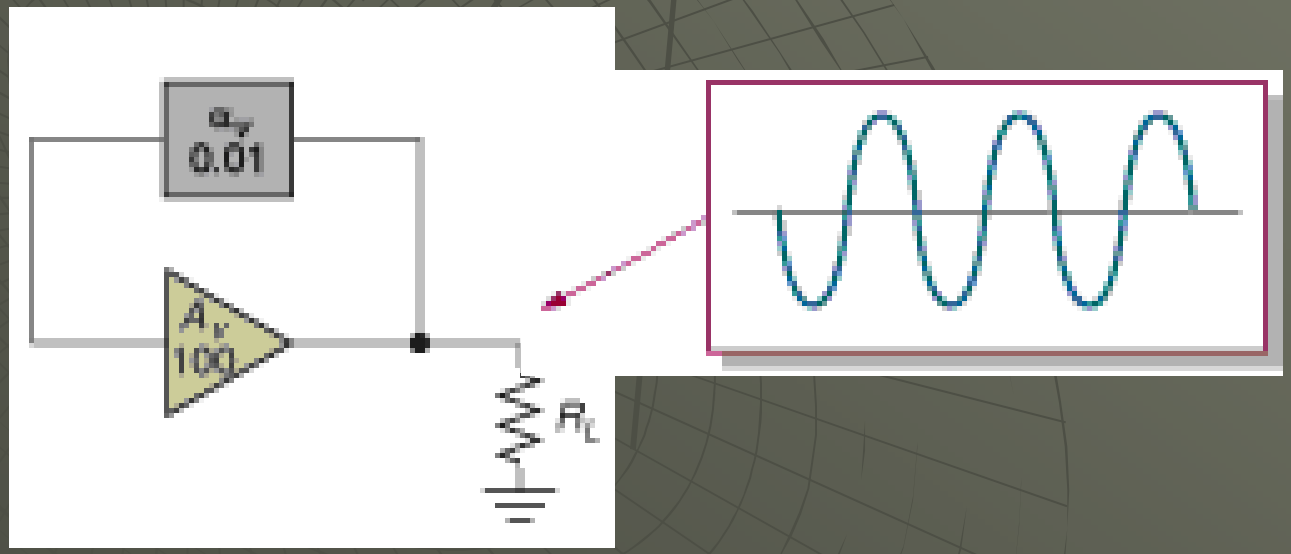
Effects of $\beta_v A_v$

If $\beta_v A_v > 1$, each oscillation results in a larger and larger signal being fed back to the input. In this case, the amplifier is quickly driven into clipping.



Effects of $\beta_v A_v$

When $\beta_v A_v = 1$, each oscillation results in a consistently equal signal being fed back to the input. One final point: Since there is always some power loss in the resistive components, in practice $\beta_v A_v$ must always be just slightly greater than 1.



Phase-Shift Oscillators

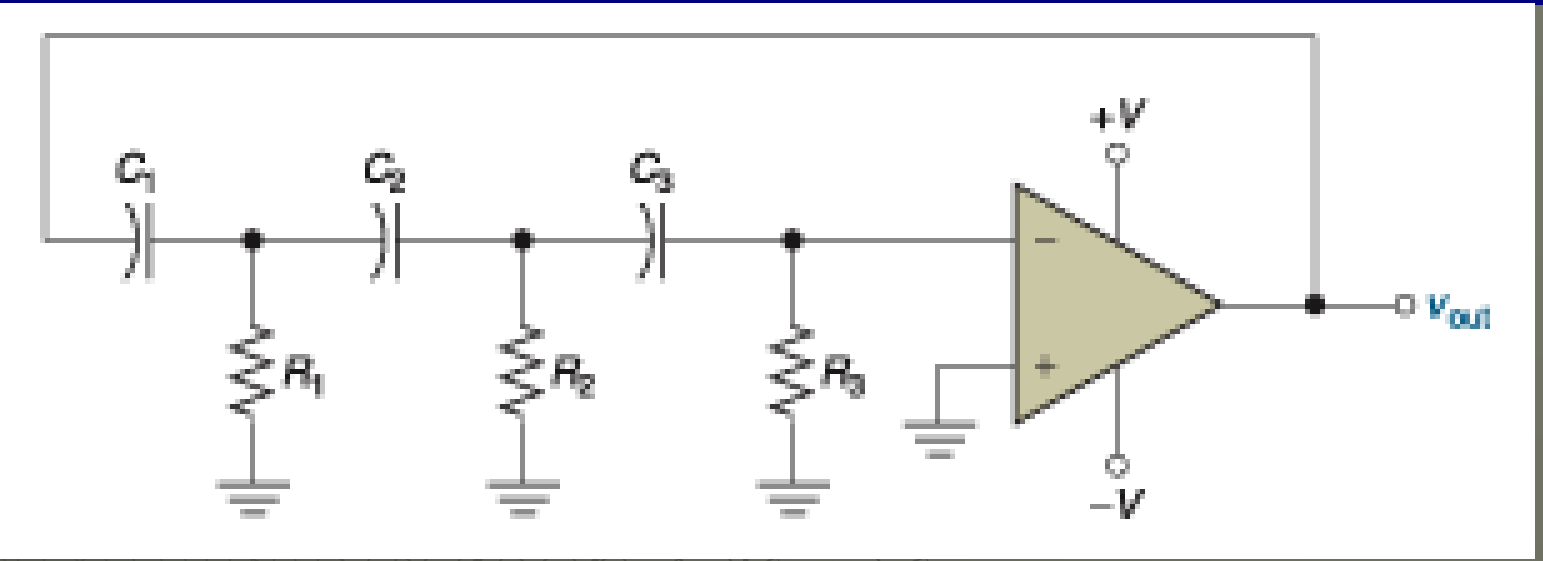
The phase-shift oscillator contains *three RC circuits* in its feedback circuit. *RC* circuit produces a phase shift at a given frequency that can be calculated using:

$$\theta = \tan^{-1}\left(\frac{-X_C}{R}\right)$$

where θ = the phase angle of the circuit

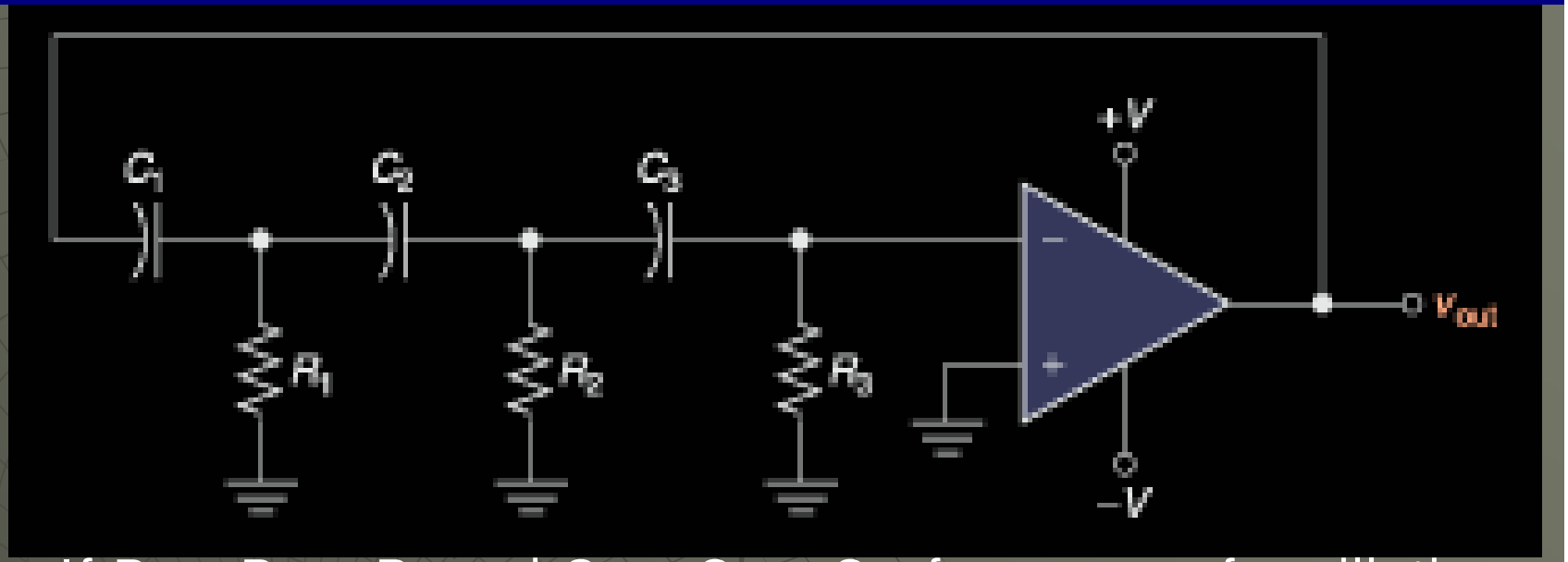
The phase angle relationship shows that θ changes with X_C and, therefore, with frequency. This means that an *RC* circuit can be designed for a specific phase shift at a given frequency.

Basic Phase-Shift Oscillators



The three *RC* networks in the phase-shift oscillator shown produce a combined phase shift of 180° at a resonant frequency (f_r). The op-amp produces another 180° phase shift, for a total circuit phase shift of 360°. As long as the Barkhausen criterion is met, the circuit oscillates at the frequency that results in the 180° phase shift.

Basic Phase-Shift Oscillators



If $R_1 = R_2 = R_3$ and $C_1 = C_2 = C_3$, frequency of oscillation will be:

$$f_r = \frac{1}{2\pi RC\sqrt{6}}$$

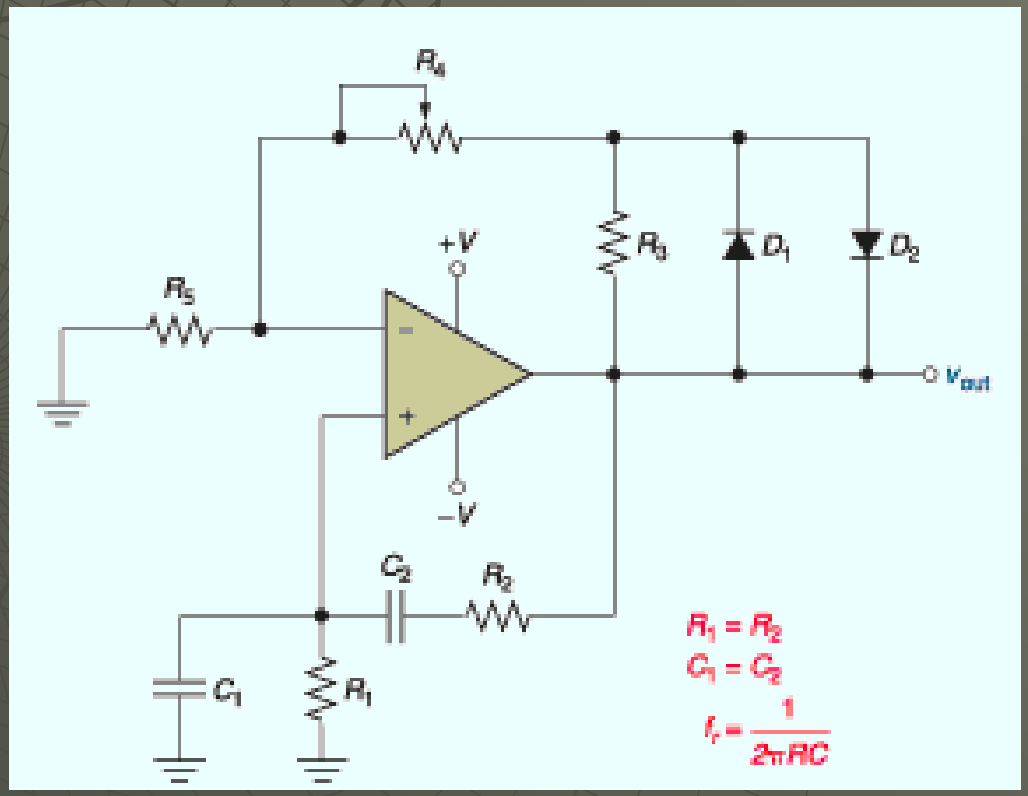
And

$$\beta_v = \frac{1}{29}$$

In reality, phase-shift oscillators are seldom used because they tend to produce outputs that are unstable in frequency and amplitude. The circuit is introduced here merely to illustrate basic oscillator operating principles.

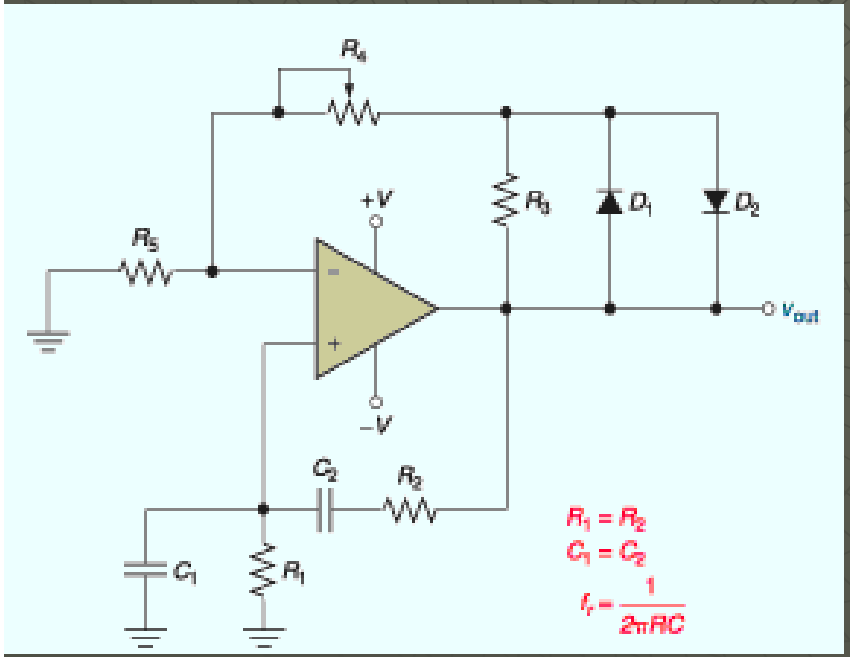
Wien-Bridge Oscillator

The *Wien-bridge oscillator* is a commonly used low-frequency oscillator. As shown in Figure, there are two *RC* circuits in the positive feedback path (output to noninverting input).



Wien-Bridge Oscillator

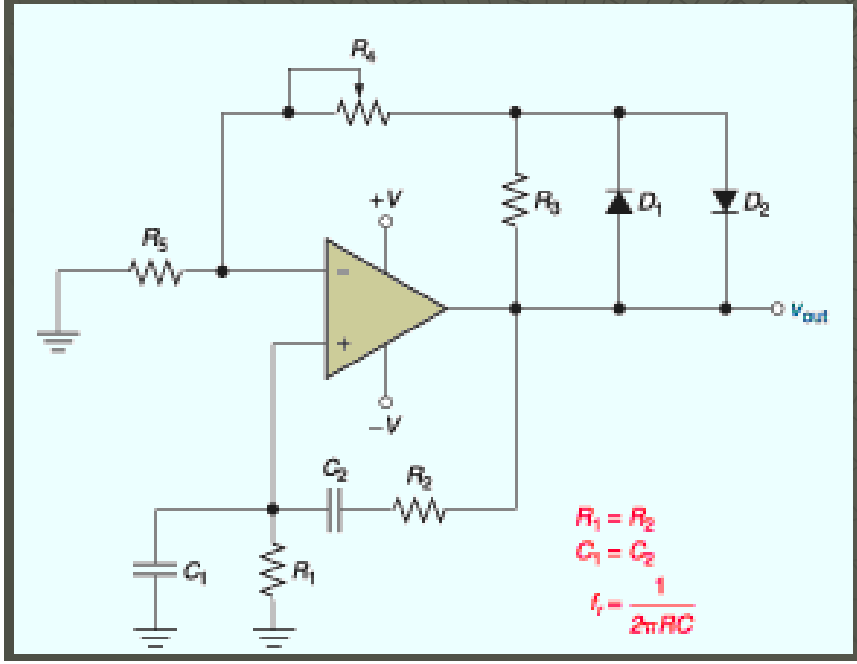
The R_1C_1 circuit forms a low-pass filter, and the R_2C_2 circuit forms a high-pass filter. Both *RC* filters have the same cutoff frequency ($R_1C_1 = R_2C_2$). Combined, they create a *band-pass filter*. As you know, a band-pass filter has *no phase shift* in its pass-band.



The circuit oscillates at the intersection of the high-pass and low-pass response curves. You may see *trimmer potentiometers* added in series with R_1 and R_2. They are used to fine-tune the circuit's operating frequency

Wien-Bridge Oscillator

The *negative feedback* path connects the output to the inverting input of the op-amp. The diodes (D_1 and D_2) limit the voltage across the feedback resistors. If the output tries to exceed the value of (V_{R4} + V_{R5}) by more than 0.7 V, one of the diodes conducts and shorts out R_3. This limits the closed-loop gain of the circuit. (The diodes are acting as *clippers* in this application.)



If $R_1 = R_2 = R$ and $C_1 = C_2 = C$, frequency of oscillation will be:

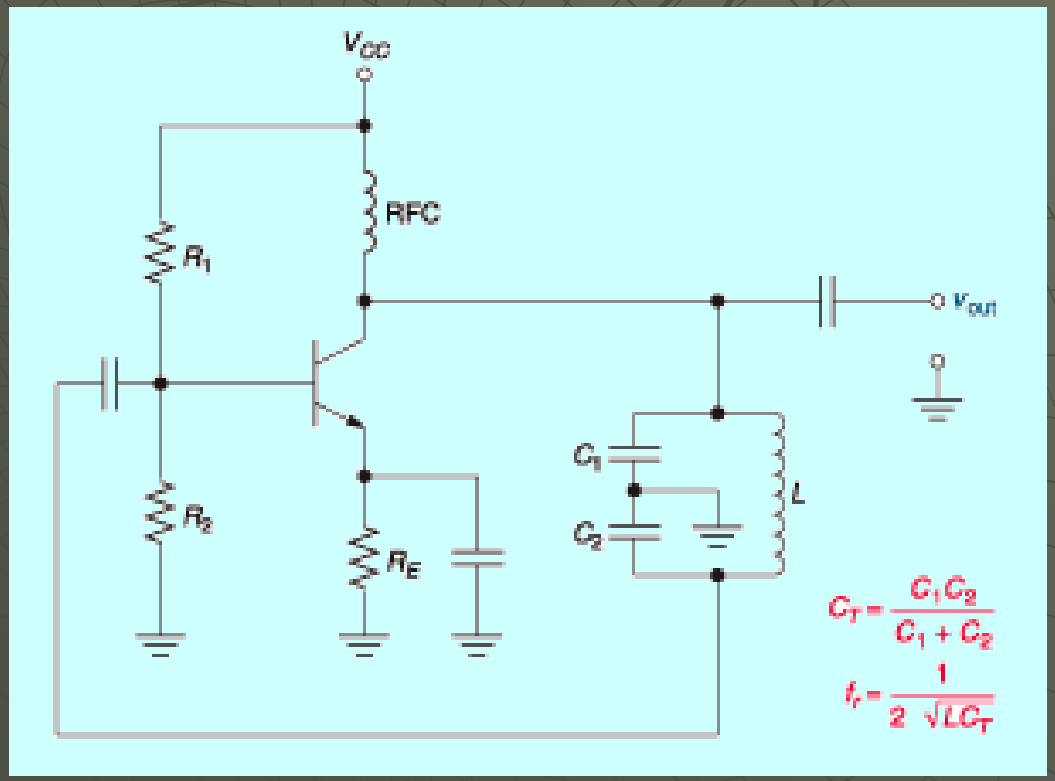
$$f_r = \frac{1}{2\pi RC}$$

And V_f

$$V_f = \frac{1}{3} V_{out}$$

Colpitts Oscillator

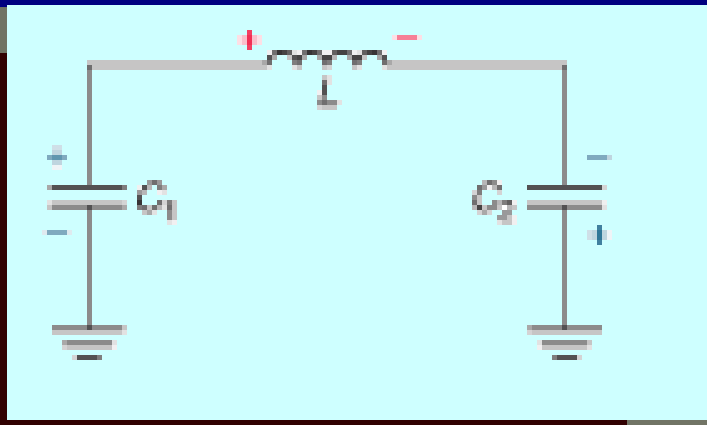
The *Colpitts oscillator* is a discrete *LC* oscillator that uses a pair of tapped capacitors and an inductor to produce regenerative feedback. The operating frequency is determined by the tank circuit. By formula:

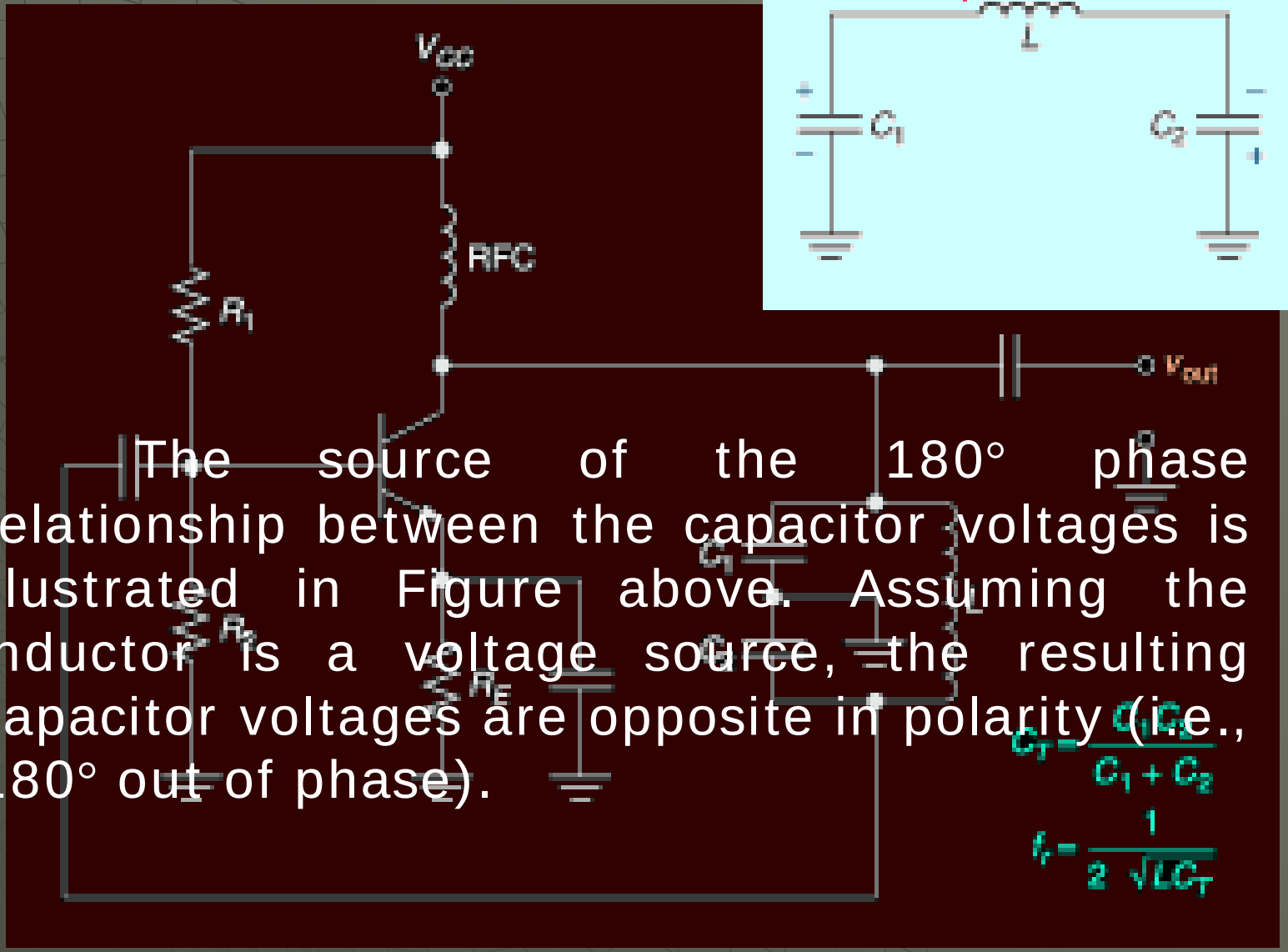


$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

$$C = \frac{C_1 C_2}{C_1 + C_2}$$

Colpitts Oscillator





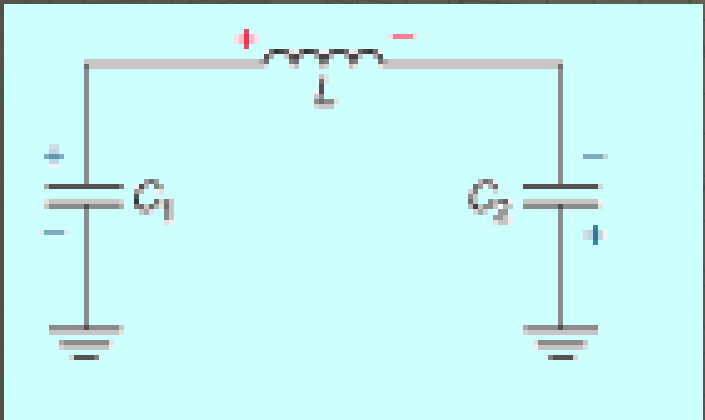
The source of the 180° phase relationship between the capacitor voltages is illustrated in Figure above. Assuming the inductor is a voltage source, the resulting capacitor voltages are opposite in polarity (i.e., 180° out of phase).

Colpitts Oscillator

The value of the feedback voltage is determined in part by the *attenuation factor* (β_v) of the feedback network. For the Colpitts oscillator

$$\beta_v = \frac{X_{C2}}{X_{C1}} = \frac{C_1}{C_2}$$

As with any oscillator, the product of $\beta_v A_v$ must be slightly greater than 1. As mentioned earlier $V_{C1} = V_{out}$ and $V_{C2} = V_f$. Therefore:



$$A_v = \frac{V_{out}}{V_f} \cong \frac{C_2}{C_1}$$

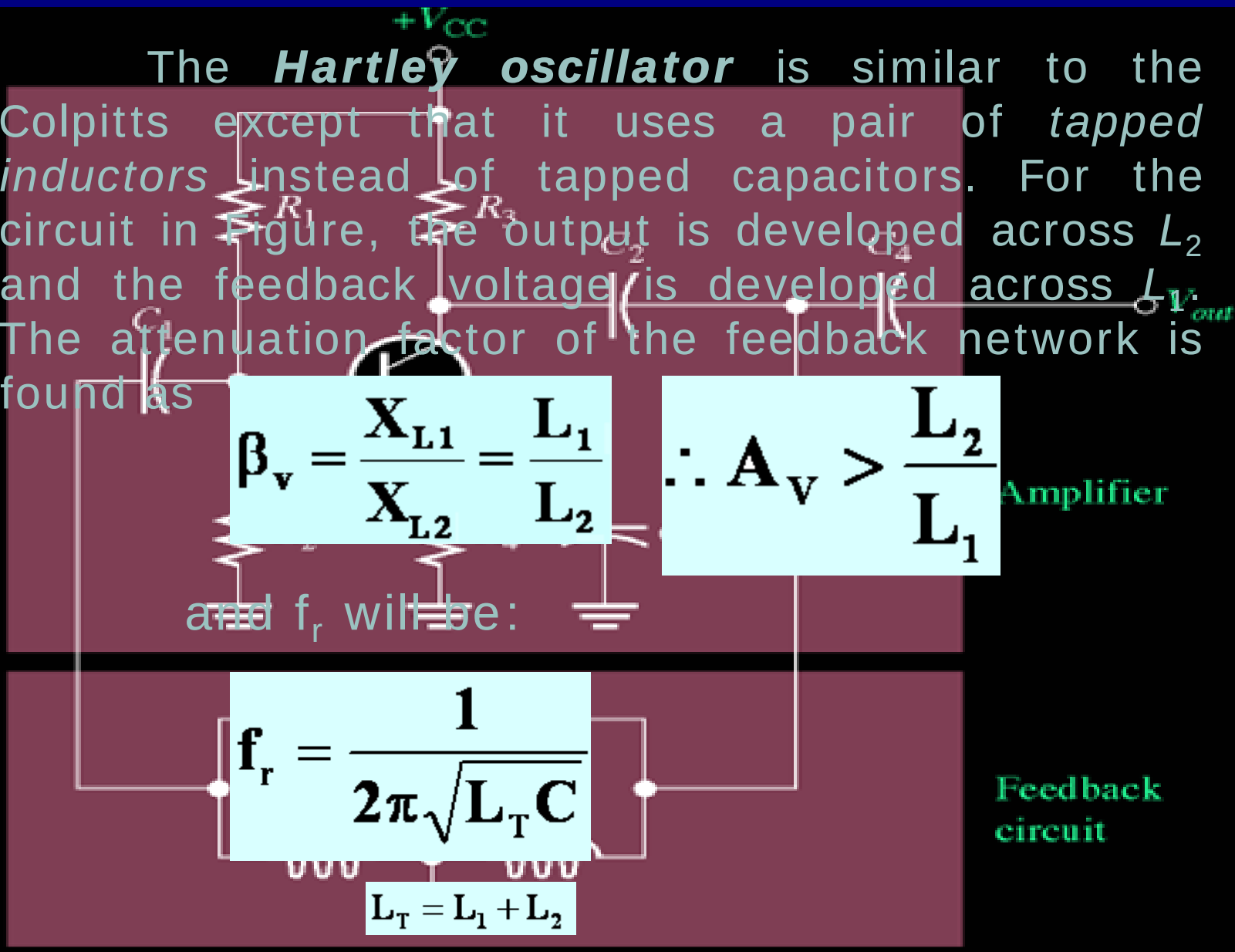
Other LC Oscillators

Three other oscillators are mentioned briefly in this lecture:

1. *Hartley oscillator*
2. *Clapp oscillator*, and
3. *Armstrong oscillator*.

Hartley Oscillator

The ***Hartley oscillator*** is similar to the Colpitts except that it uses a pair of *tapped inductors* instead of tapped capacitors. For the circuit in Figure, the output is developed across L_2 and the feedback voltage is developed across L_1. The attenuation factor of the feedback network is found as



$$\beta_v = \frac{X_{L1}}{X_{L2}} = \frac{L_1}{L_2} \qquad \therefore A_V > \frac{L_2}{L_1}$$

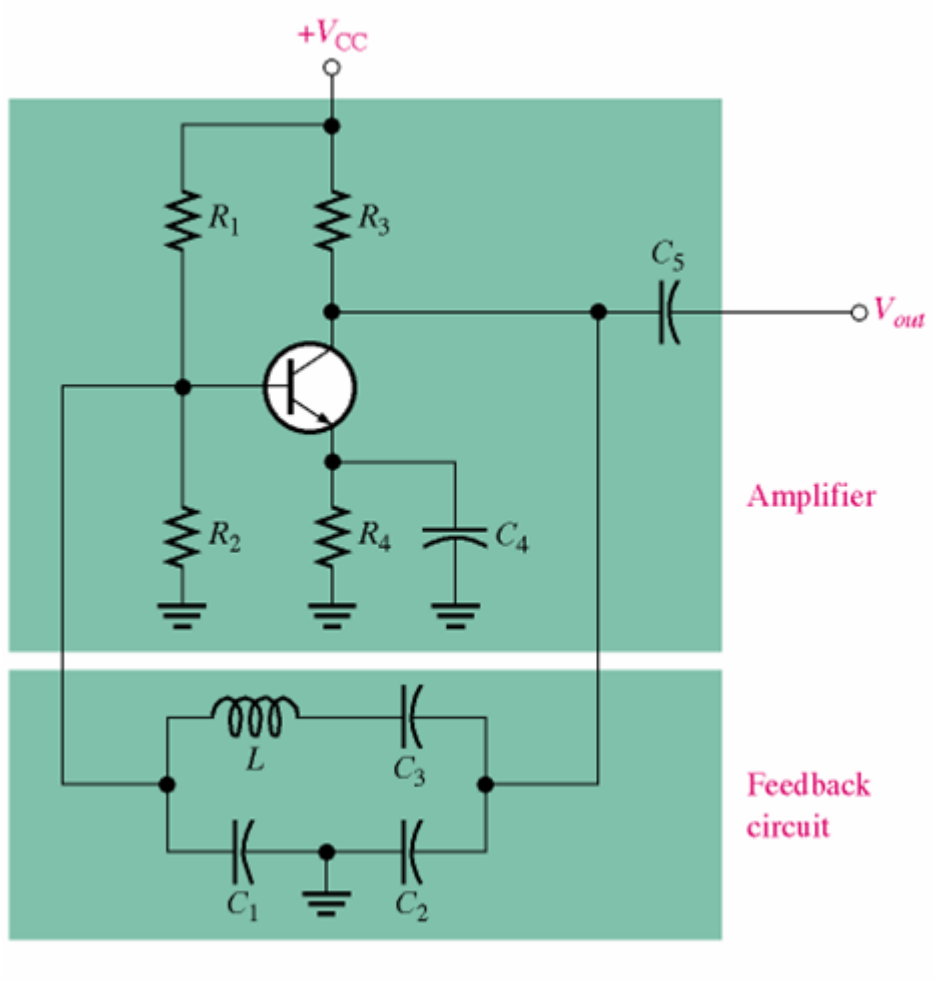
and f_r will be:

$$f_r = \frac{1}{2\pi\sqrt{L_T C}}$$

$$L_T = L_1 + L_2$$

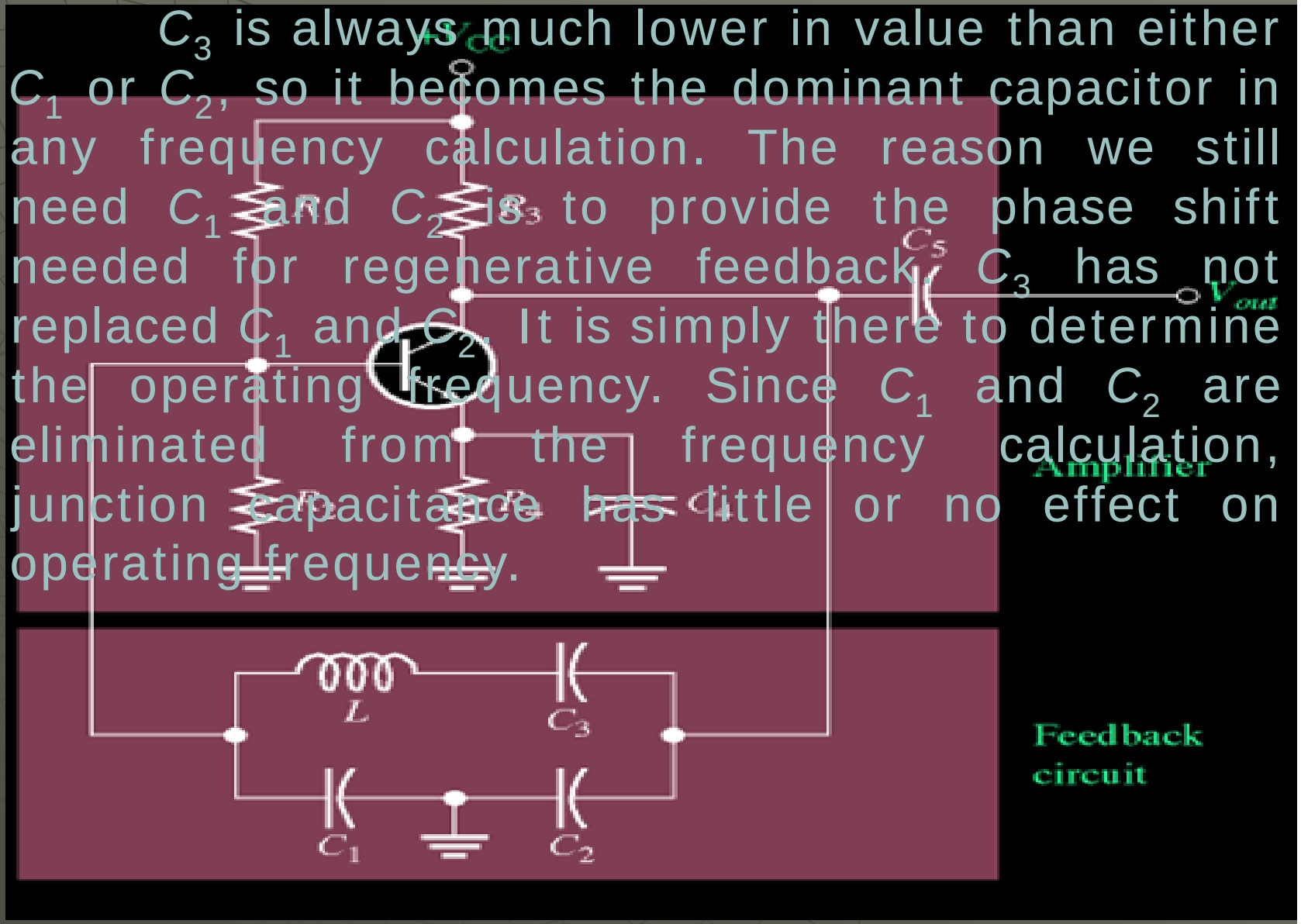
Clapp Oscillator

The ***Clapp oscillator*** is simply a Colpitts oscillator with an extra capacitor in series with the coil. It is labeled as C_3. The function of C_3 is to reduce the effects of *junction capacitance* on operating frequency.



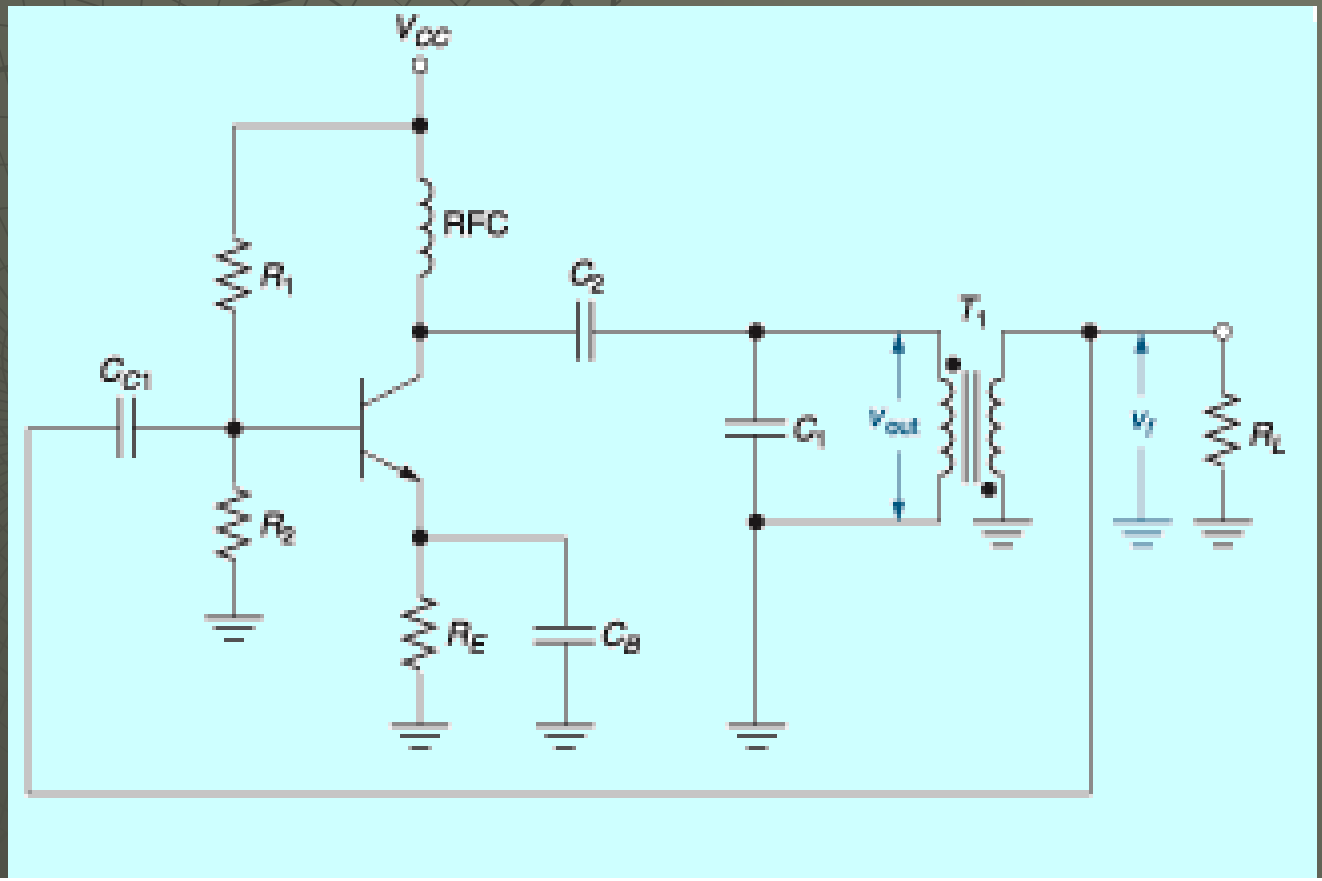
Clapp Oscillator

C_3 is always much lower in value than either C_1 or C_2, so it becomes the dominant capacitor in any frequency calculation. The reason we still need C_1 and C_2 is to provide the phase shift needed for regenerative feedback. C_3 has not replaced C_1 and C_2. It is simply there to determine the operating frequency. Since C_1 and C_2 are eliminated from the frequency calculation, junction capacitance has little or no effect on operating frequency.

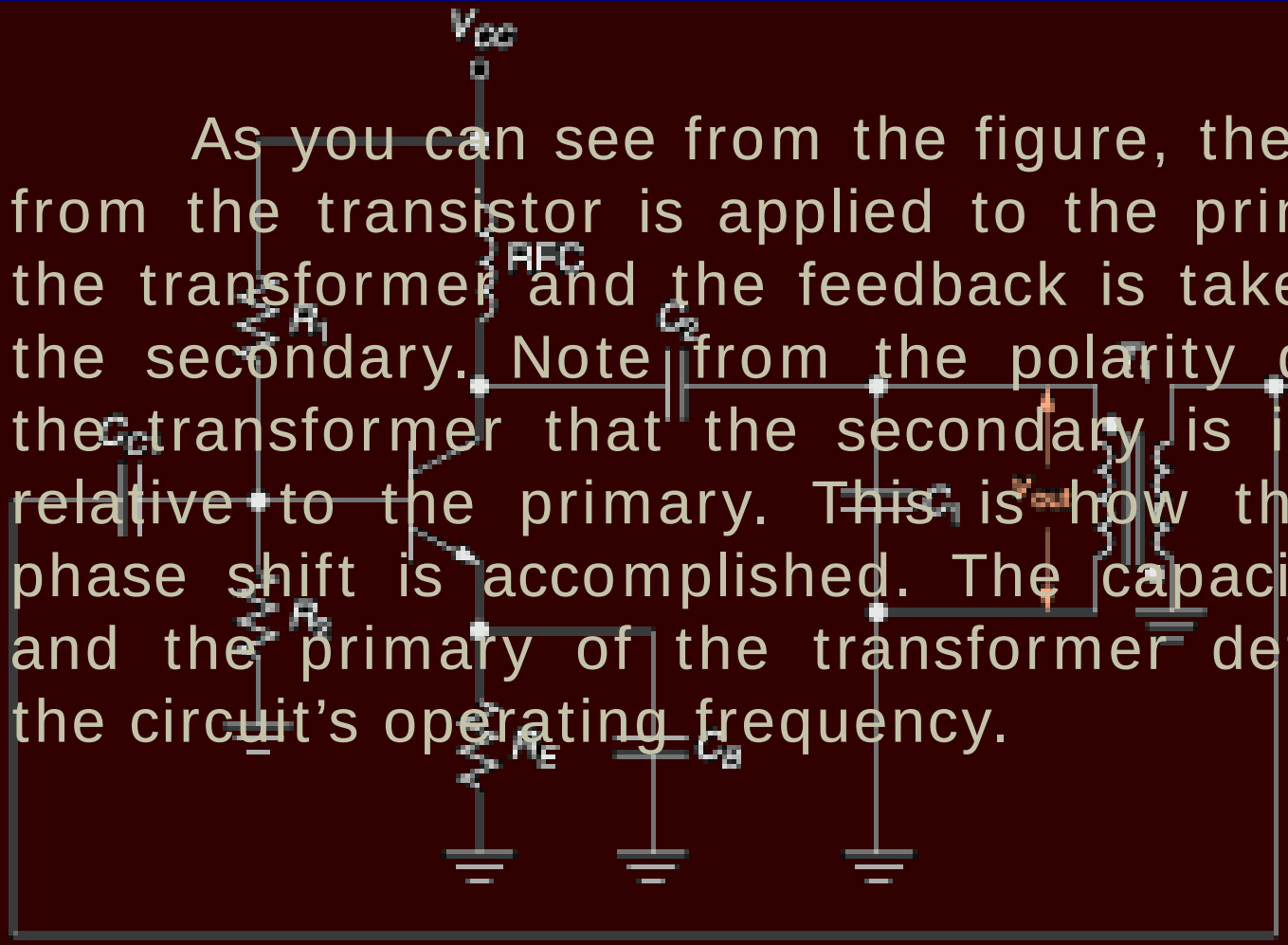


Armstrong oscillator

The ***Armstrong oscillator*** uses a transformer to achieve the 180° phase shift required for oscillation. Sometimes called as 'tickler' oscillator



Armstrong oscillator



As you can see from the figure, the output from the transistor is applied to the primary of the transformer and the feedback is taken from the secondary. Note from the polarity dots on the transformer that the secondary is inverted relative to the primary. This is how the 180° phase shift is accomplished. The capacitor, C_1, and the primary of the transformer determine the circuit's operating frequency.

RECAP

OSCILLATOR	Circuit Recognition Feature
Hartley	Tapped inductors or a tapped transformer with a single parallel capacitor.
Clapp	Looks like a Colpitts with an added capacitor in series with the coil.
Armstrong	A transformer (not tapped) in parallel with a single capacitor.
Colpitts	Tapped capacitor with a single parallel inductor.

Crystal-Controlled Oscillators

In applications where extremely stable operating frequencies are required, the oscillators described so far come up short. They can experience variations in both frequency and amplitude for several reasons:

1. If the transistor is replaced, it may have slightly different gain characteristics.
2. If the inductor or capacitor is changed, the operating frequency may change.
3. If circuit temperature changes, the resistive components will change, which can cause a change in both frequency and amplitude.

Crystal-Controlled Oscillators

In any system where stability is paramount, *crystal-controlled oscillators* are used. Crystal-controlled oscillators use a *quartz crystal* to control the operating frequency.

The key to the operation of a crystal-controlled oscillator is the *piezoelectric effect*, which means that *the crystal vibrates at a constant rate when it is exposed to an electric field*. The *physical dimensions* of the crystal determine the frequency of vibration. Thus, by cutting the crystal to specific dimensions, we can produce crystals that have very exact frequency ratings.

Crystals

There are three commonly used crystals that exhibit piezoelectric properties. They are *Rochelle salt, quartz,* and *tourmaline*. Rochelle salt has the best piezoelectric properties but is very fragile. Tourmaline is very tough, but its vibration rate is not as stable. Quartz crystals fall between the two extremes and are the most commonly used.

Quartz crystals are made from silicon dioxide (SiO_2). When used in electronic components, a thin slice of crystal is placed between two conductive plates, like those of a capacitor. Remember that its physical dimensions determine the frequency at which the crystal vibrates.

Crystals

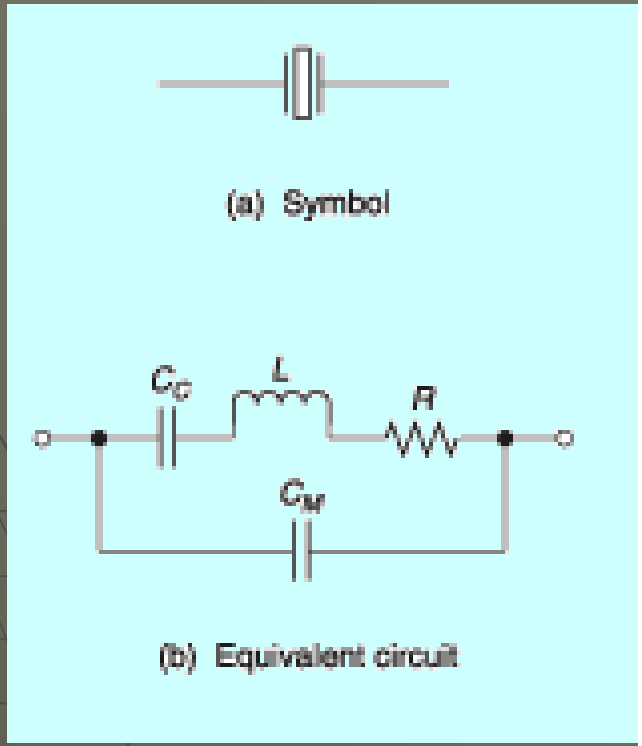
The electrical operation of the crystal is a function of its physical properties, but it can still be represented by an equivalent circuit. The equivalent circuit in figure represents specific crystal characteristics:

C_C = the capacitance of the crystal itself

C_M = the mounting capacitance, or the capacitance between the crystal and the two conducting plates

L = the inductance of the crystal

R = the resistance of the crystal

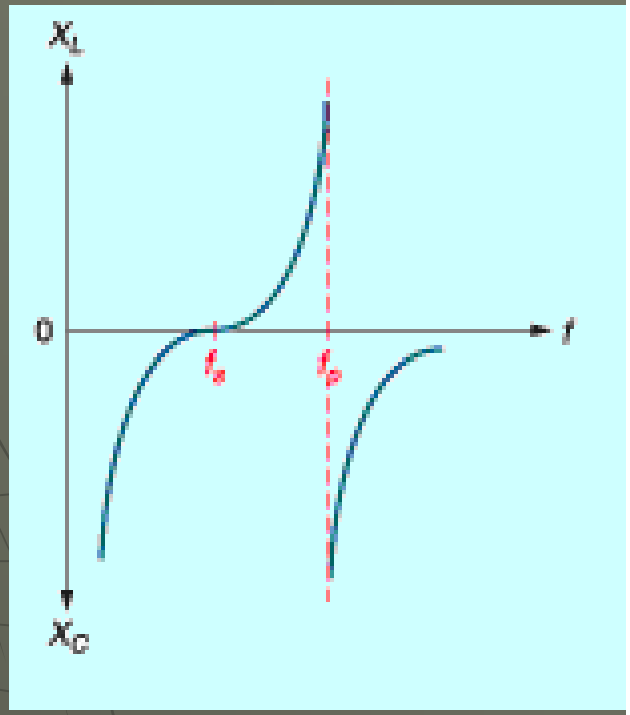


Crystals

The response curve in figure has the primary points are as follows:

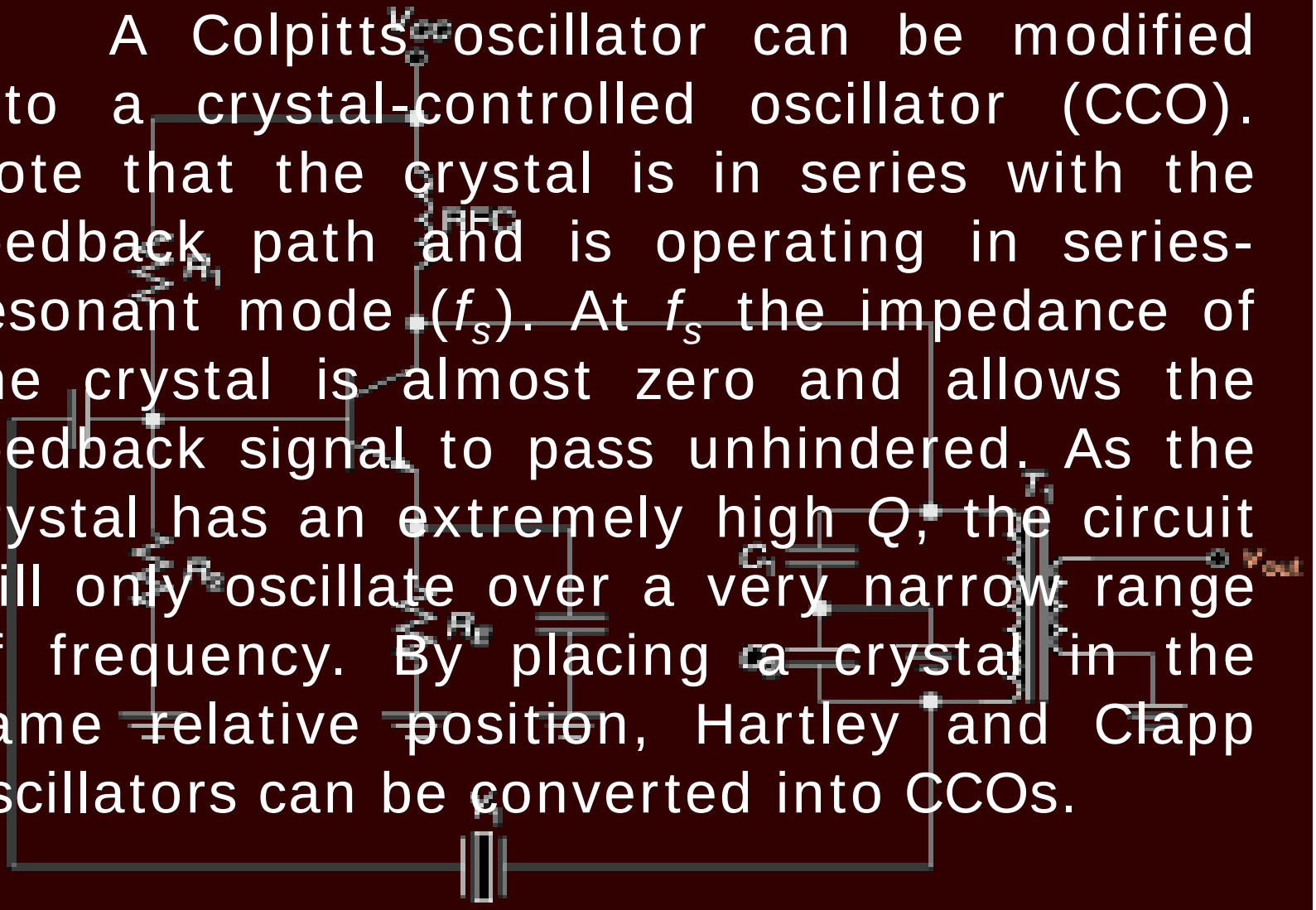
- At f_s the crystal acts as a *series resonant circuit.*
- At f_p the crystal acts as a *parallel resonant circuit.*

This means that a crystal can be used to replace either a series or a parallel resonant *LC* circuit. It should also be noted that there is very little difference between f_s and f_p.



CCO

A Colpitts oscillator can be modified into a crystal-controlled oscillator (CCO). Note that the crystal is in series with the feedback path and is operating in series-resonant mode (f_s). At f_s the impedance of the crystal is almost zero and allows the feedback signal to pass unhindered. As the crystal has an extremely high Q, the circuit will only oscillate over a very narrow range of frequency. By placing a crystal in the same relative position, Hartley and Clapp oscillators can be converted into CCOs.




THANK YOU.